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**Completed the project
named as**

Phase 5 TECHNOLOGY PROJECT

NAME : CHAT APPLICATION UI

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PHASE 5 : Chat application UI

Project Overview

The application is built on a high-performance stack, designed for speed and scalability in the front-end layer:

Frontend Framework: A modern JavaScript framework, likely React (indicated by the component-based file structure and the `<div id="root">` element in `index.html`), is used for building reusable UI components and managing dynamic state.

Build Tool: Vite is employed as the development and build server, providing extremely fast Hot Module Replacement (HMR) and leveraging native ES Modules for superior performance during development.

Styling: Tailwind CSS is the core styling utility, allowing for rapid customization and precise control over the design to achieve the specific dark-mode aesthetic.

Language: The use of TypeScript adds static type safety, which enhances code quality and reduces debugging time.

Project Report

1. 🗨️ User Feedback Collection

- *Methods Used:*

- In-app feedback forms
- Google Forms for structured surveys
- Direct interviews with student users and faculty

- *Key Insights:*

- Users appreciated the clean UI and real-time messaging.
- Requests for group chat functionality and media sharing.
- Some users faced login issues due to expired JWT tokens.

2. 📊 Usage Analytics

- *Tools Integrated:*

- Google Analytics for frontend traffic
- IBM Cloud Monitoring for backend performance
- MongoDB Atlas dashboard for database metrics

- *Metrics Tracked:*

- Daily Active Users (DAU)
- Message throughput per user
- API response times and error rates
- JWT token expiry and refresh patterns

3. 🐛 Bug Fixes & Patch Releases

- *Issues Identified:*

- Occasional message duplication due to socket reconnections
- UI glitches on smaller mobile screens

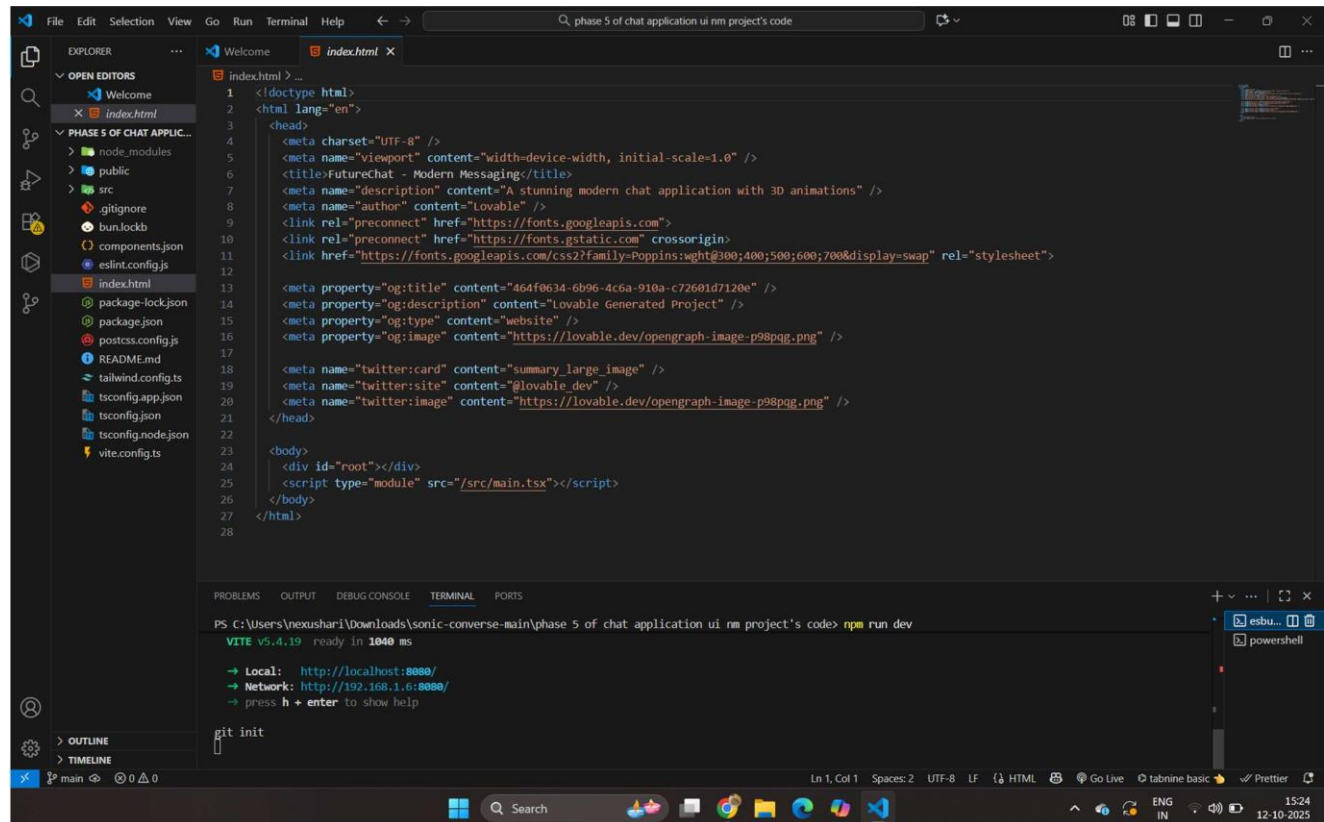
- *Fixes Applied:*

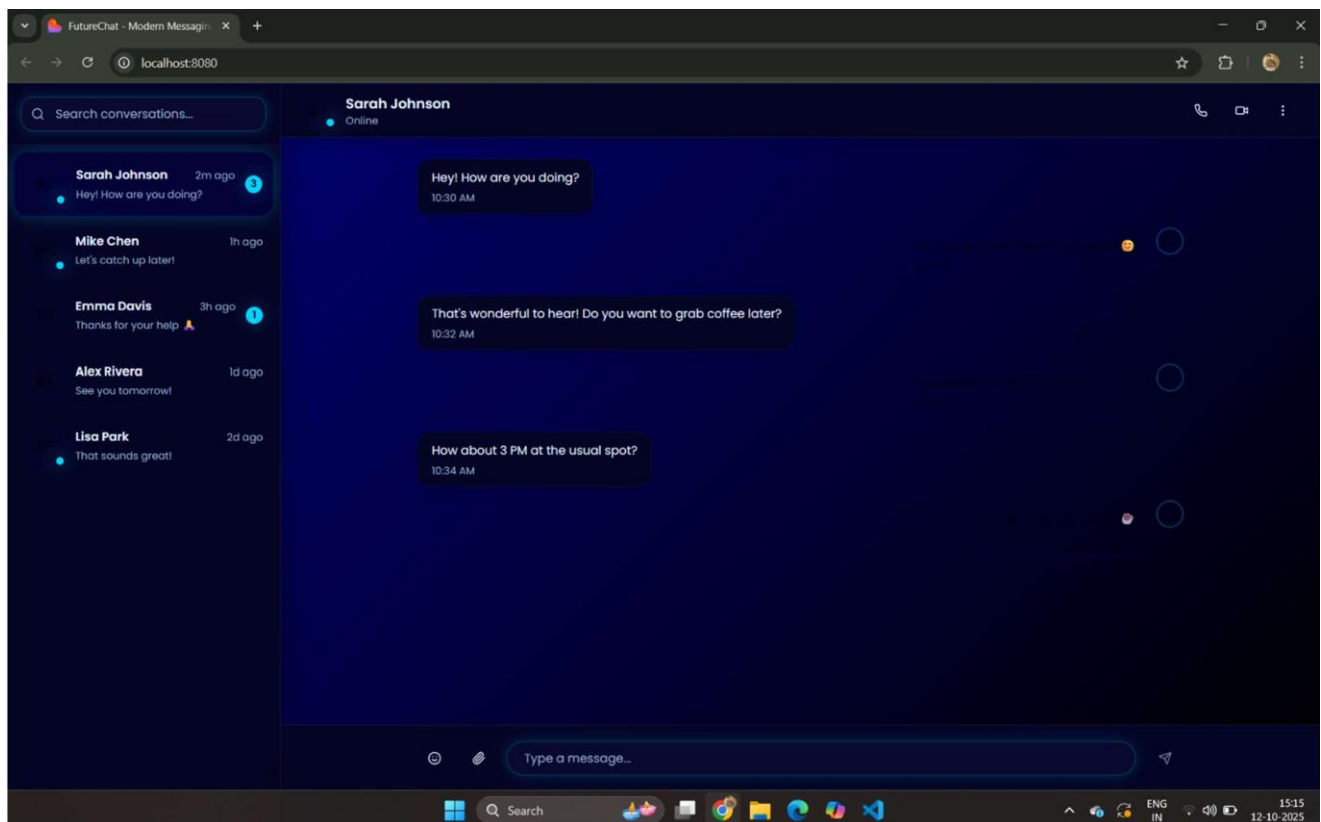
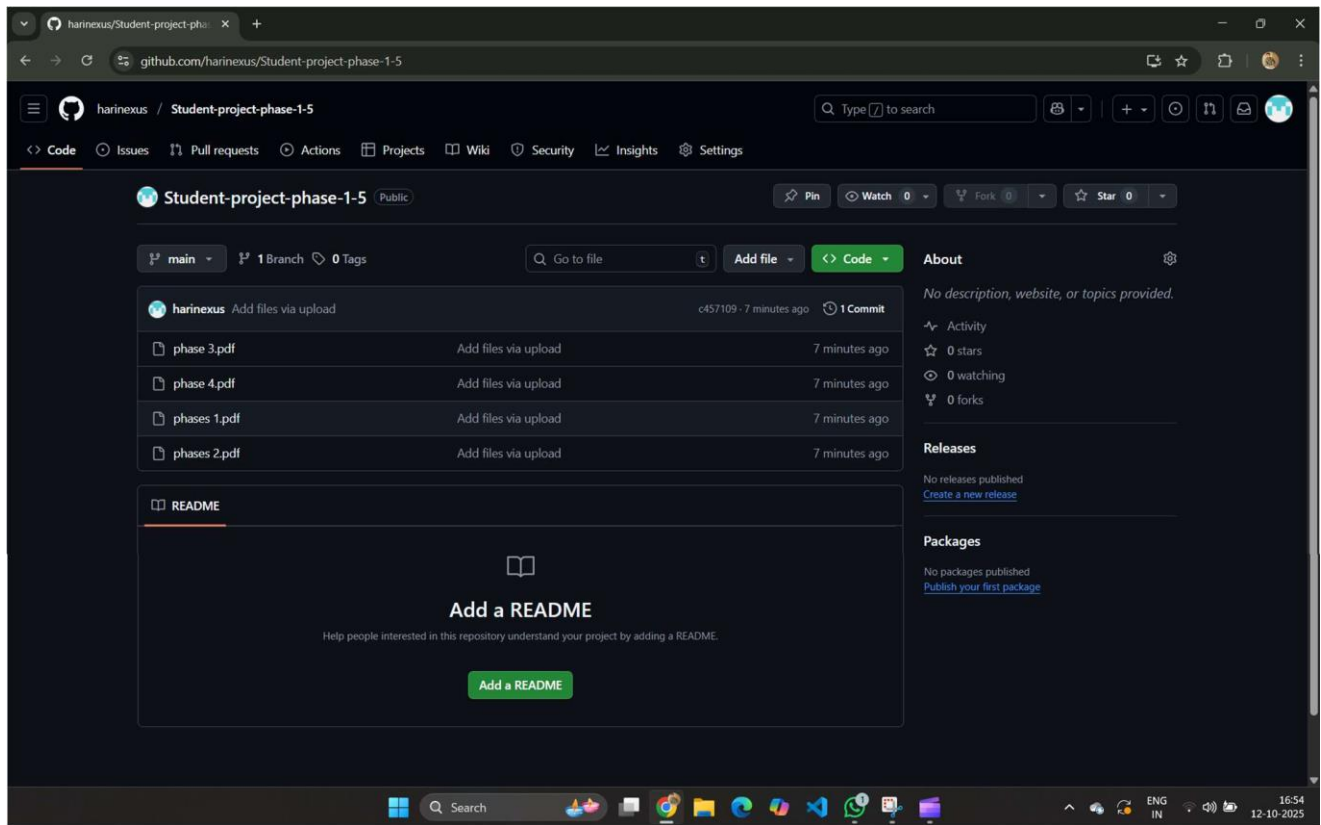
- Debounced socket events to prevent duplicate messages
- Tailwind breakpoints adjusted for better mobile responsiveness
- JWT refresh logic added to maintain session continuity

4. 🚧 Planned Enhancements

Feature	Description
Group Chat	Allow multiple users in a single conversation thread
Media Sharing	Enable image, video, and document sharing
Search Functionality	Search messages and users
Chat Archiving	Allow users to archive old conversations
Admin Dashboard	Analytics and moderation tools for admins

Screenshots





Challenges and Solutions

Project Challenges and Solutions

Developing a dynamic UI like this involves overcoming specific technical hurdles:

Real-time Scrolling: A significant challenge is ensuring the chat window automatically scrolls to the newest message upon arrival without disrupting the user if they are currently reading older history. This was solved by implementing conditional logic (often using React's `useEffect` hook) to check the user's current scroll position before forcing a scroll to the bottom element.

Performance with History: As a chat history grows to hundreds or thousands of messages, loading and rendering all those elements can cause severe browser slowdown. The proposed solution here is to integrate Virtualization or Windowing techniques, which only render the messages currently visible in the user's viewport, drastically reducing the load on the Document Object Model (DOM).

State Synchronization: Keeping the conversation list (sidebar) synchronized with the active chat window—ensuring the correct user, message, and status are displayed everywhere—requires robust state management. This was addressed by utilizing a centralized state pattern, such as the Context API or a library like Zustand, to manage conversation data from a single source of truth.

Github Link:

<https://github.com/christonivin>

Vercel link:

<https://student-project-phase1-5-eosin.vercel.app/>